

Systems of Linear Differential Equations

7.1 First-Order Linear Systems of Differential Equations

Standard Form

A first-order linear system of differential equations in standard form is given by:

$$\begin{cases} y_1'(t) = a_{11}(t)y_1(t) + a_{12}(t)y_2(t) + \cdots + a_{1n}(t)y_n(t) + g_1(t) \\ y_2'(t) = a_{21}(t)y_1(t) + a_{22}(t)y_2(t) + \cdots + a_{2n}(t)y_n(t) + g_2(t) \\ \vdots \\ y_n'(t) = a_{n1}(t)y_1(t) + a_{n2}(t)y_2(t) + \cdots + a_{nn}(t)y_n(t) + g_n(t) \end{cases}$$

This can be written in matrix form as:

$$\begin{pmatrix} y_1'(t) \\ y_2'(t) \\ \vdots \\ y_n'(t) \end{pmatrix} = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{pmatrix} \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix} + \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix}$$

In compact vector notation:

$$\mathbf{y}'(t) = A(t)\mathbf{y}(t) + \mathbf{g}(t).$$

- $A(t)$ is the **coefficient matrix**.
- $\mathbf{y}(t)$ is the vector of unknown functions.
- $\mathbf{g}(t)$ is a given forcing term.

Classification

- **Homogeneous:** If $\mathbf{g}(t) = \mathbf{0}$, the system is homogeneous: $\mathbf{y}'(t) = A(t)\mathbf{y}(t)$.
- **Constant Coefficient:** If the entries of $A(t)$ are constants, the system has constant coefficients.

Focus of Course: Homogeneous first-order linear systems with constant coefficients:

$$\mathbf{y}'(t) = A\mathbf{y}(t), \quad \text{where } A \in \mathbb{R}^{n \times n}.$$

Solutions

- A function-valued vector $\mathbf{x}(t)$ is a **solution** to the system if it satisfies the differential equation.
- An **initial condition** specifies the value of the solution at some initial time t_0 : $\mathbf{y}(t_0) = \mathbf{a}$, where $\mathbf{a} \in \mathbb{R}^n$.
- An **initial value problem (IVP)** consists of a differential equation together with an initial condition.

Theorem: Linear Combination of Solutions

If $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are solutions to $\mathbf{y}' = A\mathbf{y}$, then for any scalars $\alpha, \beta \in \mathbb{R}$, the linear combination $\alpha\mathbf{x}_1(t) + \beta\mathbf{x}_2(t)$ is also a solution. (Superposition principle)

Linear Independence of Solutions

- A set $\{\mathbf{x}_1(t), \mathbf{x}_2(t), \dots, \mathbf{x}_k(t)\}$ of function-valued vector is linearly independent if whenever $c_1, c_2, \dots, c_k \in \mathbb{R}$ are real numbers such that $c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) + \cdots + c_k\mathbf{x}_k(t) = \mathbf{0}$ for all t in the domain, necessarily $c_1 = c_2 = \cdots = c_k = 0$. Otherwise, we say that it is linearly dependent.

Wronskian

Let $S = \{\mathbf{x}_1(t), \mathbf{x}_2(t), \dots, \mathbf{x}_n(t)\}$ be a set of n function-valued vectors with n coordinates. The **Wronskian** of S is defined to be the determinant of the matrix formed by these vectors:

$$W(\mathbf{x}_1(t), \mathbf{x}_2(t), \dots, \mathbf{x}_n(t)) = \det \begin{pmatrix} \mathbf{x}_1(t) & \mathbf{x}_2(t) & \cdots & \mathbf{x}_n(t) \end{pmatrix}.$$

- If $W(\mathbf{x}_1(t), \mathbf{x}_2(t), \dots, \mathbf{x}_n(t)) \neq 0$, then the set S is linearly independent.

Fundamental Set of Solutions

- A set $S = \{\mathbf{x}_1(t), \mathbf{x}_2(t), \dots, \mathbf{x}_n(t)\}$ of solutions to the differential system $\mathbf{y}' = A\mathbf{y}$ is called a **fundamental set of solutions** if its Wronskian is nonzero.
- The **general solution** to $\mathbf{y}' = A\mathbf{y}$ is a linear combination of the function-valued vectors in a fundamental set of solutions:

$$\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) + \dots + c_n\mathbf{x}_n(t), \quad c_i \in \mathbb{R}.$$

Theorem: Solutions for Diagonalizable Coefficient Matrices

Suppose A is diagonalizable. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be n linearly independent eigenvectors associated to eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (not necessarily distinct). Then:

- The set $\{\mathbf{v}_1 e^{\lambda_1 t}, \mathbf{v}_2 e^{\lambda_2 t}, \dots, \mathbf{v}_n e^{\lambda_n t}\}$ is a fundamental set of solutions to the first-order homogeneous system $\mathbf{y}' = A\mathbf{y}$.
- The general solution is

$$\mathbf{x}(t) = c_1\mathbf{v}_1 e^{\lambda_1 t} + c_2\mathbf{v}_2 e^{\lambda_2 t} + \dots + c_n\mathbf{v}_n e^{\lambda_n t}, \quad c_i \in \mathbb{R}.$$

7.2 Complex Eigenvalues and Eigenvectors

Complex Vectors

An n -complex vector is a collection of n ordered complex numbers. Denoted as $\mathbb{C}^n$. A complex vector

can be split into real and imaginary parts. If $\mathbf{v} = \begin{pmatrix} x_1 + iy_1 \\ x_2 + iy_2 \\ \vdots \\ x_n + iy_n \end{pmatrix}$, then $\mathbf{v} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + i \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \text{Re}(\mathbf{v}) + i\text{Im}(\mathbf{v})$.

Complex Eigenvalues and Eigenvectors

- A complex number $\lambda \in \mathbb{C}$ is an eigenvalue of a real matrix A if there is a nonzero vector $\mathbf{v} \in \mathbb{C}^n$ such that $A\mathbf{v} = \lambda\mathbf{v}$. In this case, $\mathbf{v}$ is called an eigenvector associated to λ .
- **Theorem:** Complex eigenvalues of a real matrix come in conjugate pairs: if λ is a complex eigenvalue of A , then $\bar{\lambda}$ is also an eigenvalue of A . Furthermore, if $\mathbf{v}$ is an eigenvector associated to λ , then $\bar{\mathbf{v}}$ is an eigenvector associated to $\bar{\lambda}$.

Real Solutions from Complex Solutions

Let $\lambda = \alpha + i\beta$ be a complex eigenvalue of a real matrix A , and let $\mathbf{v} = \mathbf{u} + i\mathbf{w}$ be a corresponding eigenvector, where $\alpha, \beta \in \mathbb{R}$ and $\mathbf{u}, \mathbf{w} \in \mathbb{R}^n$.

- Real Solutions

$$x_1(t) = e^{\alpha t}(\cos(\beta t)\mathbf{u} - \sin(\beta t)\mathbf{w}) \quad x_2(t) = e^{\alpha t}(\sin(\beta t)\mathbf{u} + \cos(\beta t)\mathbf{w})$$

Fundamental Set and General Solutions with Complex Eigenvalues

Then, assuming λ has multiplicity one and we have no repeated complex eigenvalues:

1. The above solutions $x_1(t), x_2(t)$ are real and are linearly independent.
2. These vectors form a linearly independent set of solutions $\{\mathbf{x}_1(t), \mathbf{x}_2(t)\}$ to differential equations $\mathbf{y}' = A\mathbf{y}$. Any solution to $\mathbf{y}' = A\mathbf{y}$ may be expressed in terms of these vectors.

$$\mathbf{y} = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) = c_1 e^{\alpha t}(\cos(\beta t)\mathbf{u} - \sin(\beta t)\mathbf{w}) + c_2 e^{\alpha t}(\sin(\beta t)\mathbf{u} + \cos(\beta t)\mathbf{w})$$

7.3 Repeated Eigenvalues and Generalized Eigenvectors

Repeated Eigenvalue

Let A be an $n \times n$ square matrix. An eigenvalue λ of A is a **repeated eigenvalue** if its algebraic multiplicity r_λ is greater than 1.

Suppose the algebraic multiplicity is strictly larger than the geometric multiplicity

$$\dim(E_\lambda) < r_\lambda$$

In this case, we may not have linearly independent eigenvectors associated to eigenvalue λ to diagonalizable A , and thus may not have enough fundamental set of solutions.

Generalized Eigenvector

1. Let $\mathbf{v}_1$ be an eigenvector associated to λ .
2. A vector $\mathbf{v}_2$ is a **generalized eigenvector** associated to λ if it is a solution to $(A - \lambda I)\mathbf{x} = \mathbf{v}_1$.

Solutions from Repeated Eigenvalue and Generalized Eigenvector

Theorem: Let A be an order 2 square matrix with a repeated eigenvalue λ . Suppose $\mathbf{v}_1$ is an eigenvector, and $\mathbf{v}_2$ a generalized eigenvector associated to λ . Then the general solution to $\mathbf{y}' = A\mathbf{y}$

$$\mathbf{x}(t) = c_1 e^{\lambda t} \mathbf{v}_1 + c_2 e^{\lambda t} (t \mathbf{v}_1 + \mathbf{v}_2).$$

- The set $\mathbf{v}_1 e^{\lambda t}, e^{\lambda t}(t \mathbf{v}_1 + \mathbf{v}_2)$ is a fundamental set of solutions to the differential equations $\mathbf{y}' = A\mathbf{y}$ for the 2×2 matrix A

Algorithm

- Find repeated eigenvector and generalized eigenvector.
- Form a generalized matrix A by applying formula
*The multiplicity of a eigenvector to the given transformation cannot be higher the size of that eigenvector.